# Automated Light System

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# Why Does Light Matter?

### Stops Melatonin Production
Signals daytime to the brain to promote alertness

### Sets Circadian Rhythm
Synchronizes internal biological clock with daily natural cycles.

### Improves Mood
Triggers serotonin release to boost energy, focus, and overall well-being.

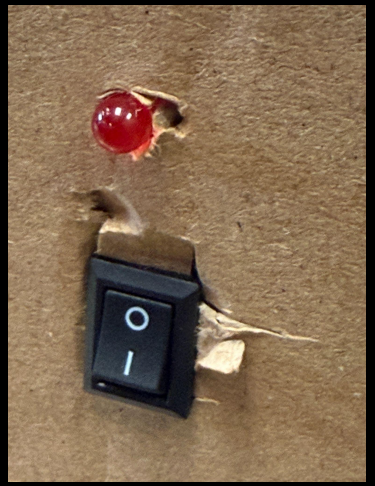


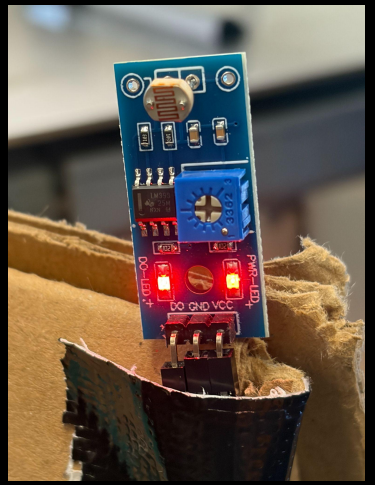
# Components

- Daylight sensor
- Humidity sensor
- Servo
- On/off switches
- LEDs

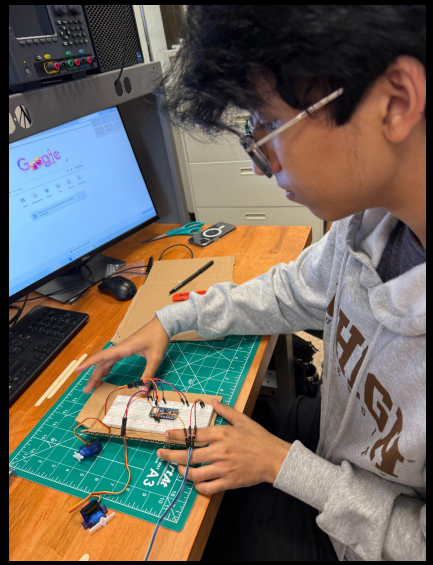


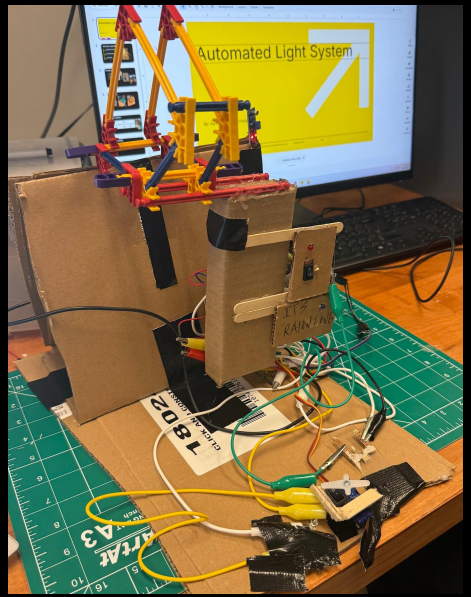






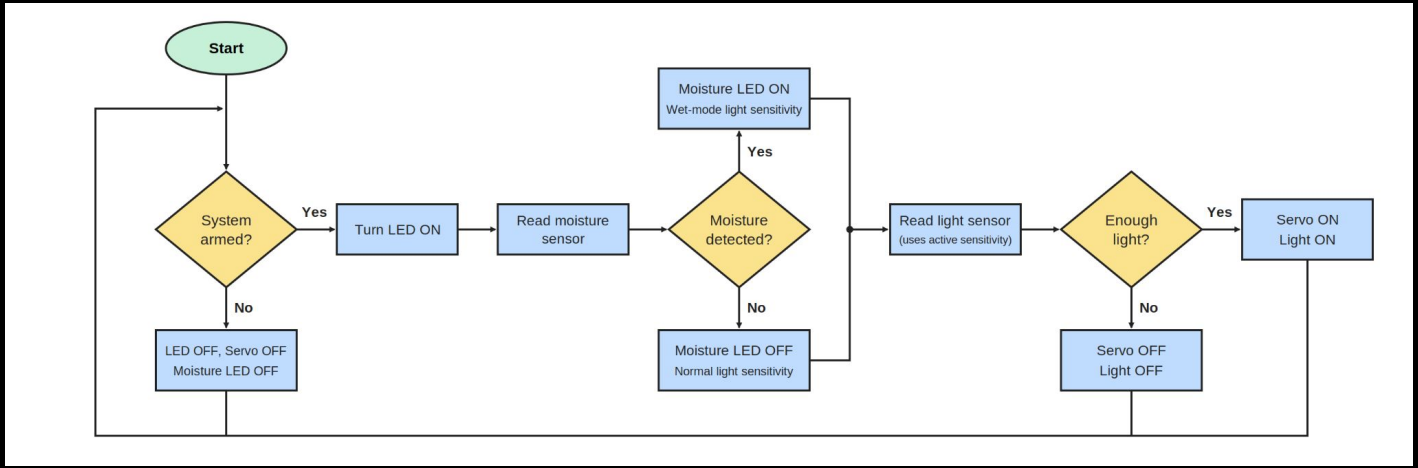
# Prototyping

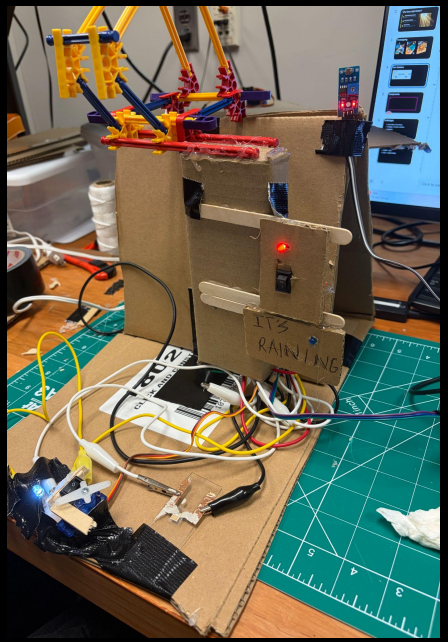







# Our Solution:



# Challenges

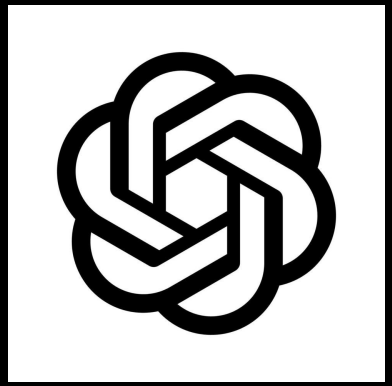


- Implementation of Servo Motors
- Structural Stability
- Hardware problems
- Wire Organization

# AI Use

- ChatGPT → Code (Software problems)
- Claude → Diagram & Wiring





# Conclusion/Results



- Working prototype with some flaws
- Improvements:
  - Better rain sensor system
  - Finalize the mounting design
  - Improve the wire management



# Thank You!